ST 4035 – Data Science

Assignment #1 – Kaggle Competition

Employee Attrition Prediction

Overview

A mid-sized technology company is experiencing unexpectedly high employee turnover. The HR department wants to proactively identify employees who are at risk of leaving (i.e., attrition) so that targeted retention strategies can be applied. You are provided with a simulated dataset of 3,500 employee records and 19 input features. Your task is to build a Logistic Regression classification model to predict the binary target variable **Attrition**.

Tasks

1. Load the dataset and perform basic Exploratory Data Analysis (EDA). Include summary statistics, class distribution, and at least two meaningful visualisations.
2. Preprocess the data: handle categorical variables (e.g., encoding), scale or normalise numeric features, and check for missing values.
3. Train a Logistic Regression model on the training dataset. Do not use regularisation, ensemble methods, or any other classification algorithm.
4. Generate predictions for the test dataset (employee_test.csv) and submit them in the sample submission format to Kaggle.

Dataset Features

The training file `employee_train.csv` contains 3,500 records with the following columns:

Feature	Description
`EmployeeID`	Unique identifier for each employee
`Age`	Employee age (18–60)
`Gender`	Male / Female
`Department`	Sales, Research & Development, Human Resources
`JobRole`	Specific role within the department
`YearsAtCompany`	Total years employed at the company
`MonthlyIncome`	Monthly salary in USD
`DistanceFromHome`	Distance from home to office (km)
`NumCompaniesWorked`	Number of companies previously worked at
`OverTime`	Whether the employee works overtime (Yes/No)
`TrainingTimesLastYear`	Number of training sessions attended last year
`WorkLifeBalance`	Self-rated work-life balance (1–4)
`JobSatisfaction`	Self-rated job satisfaction (1–4)
`EnvironmentSatisfaction`	Satisfaction with work environment (1–4)

`RelationshipSatisfaction`	Satisfaction with workplace relationships (1–4)
`JobInvolvement`	Level of job involvement (1–4)
`PerformanceRating`	Last performance review rating (3–4)
`YearsSinceLastPromotion`	Years since the last promotion
`YearsWithCurrManager`	Years under current manager
`BusinessTravel`	Travel frequency: Non-Travel, Travel_Rarely, Travel_Frequently
`Attrition`	Target variable — 1 = Left company, 0 = Stayed (train only)

Files Provided

- `employee_train.csv` — 3,500 labelled training records (includes Attrition column)
- `employee_test.csv` — 1,500 unlabelled test records (no Attrition column)
- `sample_submission.csv` — Template for Kaggle submission (EmployeeID + Attrition)

Submissions

5. Submit your report as a PDF to the LMS. The report must include your R/Python code, EDA outputs, preprocessing steps, model results, and interpretations for Tasks 1–3.
6. Upload your predictions file in the sample submission format to Kaggle. Use your real name when creating your Kaggle account.

Kaggle Competition Link

https://www.kaggle.com/t/c0e05cab6f394cb99769040d8ce8c240

Grading Criteria

Component	Marks	Total
Exploratory Data Analysis (Task 1)	25	
– Summary statistics & class distribution	10	
– Visualisations (at least 2)	15	
Data Preprocessing (Task 2)	25	
– Encoding & scaling	15	
– Missing value check & justification	10	
Logistic Regression Model (Task 3)	30	
– Model training & evaluation metrics	20	
– Coefficient interpretation	10	
Kaggle Submission & Predictions (Task 4)	20	
Total		**100**

Please use your real name when creating your Kaggle account.